

KL-Constrained Baum-Welch for Order-Misspecified HMMs:

A Null Result from a Stable Implementation

Anonymous Authors

Abstract

Hidden Markov models (HMMs) are routinely fitted with first-order Baum-Welch to data generated by higher-order processes, inducing irreducible structural bias. KL-Constrained Baum-Welch (KL-BW) modifies each M-step transition-row update by projecting onto a KL ball of radius ε around the previous iterate, motivated by proximal EM theory. Prior experimental iterations reported that KL-BW yielded catastrophic accuracy collapses at $d = 16$ states, driven by single-seed failures in an implementation based on Lagrange multiplier binary search. We revisit this claim with a fully reproducible experiment (315 configurations: 5 seeds $\times$ 3 Markov orders $\times$ 3 state-space sizes $\times$ 7 sequence lengths) using a numerically stable geometric-interpolation M-step. We further introduce the BW-warm ablation—standard Baum-Welch with the same extra 30 iterations from the same warm initialization as KL-BW—to isolate constraint effects from iteration effects. The main finding is a null result: across all 315 configurations, Δ_2 (KL-BW minus BW-warm accuracy) lies in $[-0.15, +0.21]$ percentage points, with 95% bootstrap confidence intervals crossing zero in 8 of 9 (k_{true}, d) cells. The constraint is active (mean constraint activation rate 40–58%) but yields no systematic benefit or harm. Prior catastrophic failures at $d = 16$ do not replicate with the stable implementation. Bias-variance decomposition confirms that structural misspecification creates large irreducible error floors (76% at $k_{\text{true}} = 2$, 84% at $k_{\text{true}} = 3$) that are identical across all three methods, reinforcing the null result.

1 Introduction

Hidden Markov models are among the most widely deployed sequence models in computational biology, speech processing, and natural language processing [Rabiner, 1989]. The standard learning algorithm—Baum-Welch, the expectation-maximization (EM) procedure for HMMs [Baum et al., 1970]—converges to local maxima of the observed-data likelihood. In practice, the true Markov order of the hidden process is rarely known, and practitioners routinely fit first-order HMMs to data with second- or higher-order memory.

Order misspecification introduces a fundamental challenge distinct from state-count misspecification: the fitted model class cannot contain the data-generating process regardless of sample size, creating irreducible structural bias. Under order mismatch, the Fisher Information Matrix of the fitted model can become singular [Dwivedi et al., 2020], degrading EM convergence rates from the parametric $O(1/\sqrt{n})$ to slower dimension-dependent rates. Douc et al. [2011], Douc and Moulines [2012] establish that the MLE in misspecified HMMs converges to the KL-divergence minimizer over the fitted class—the *pseudo-true* parameter—but do not characterize the EM convergence path.

A natural algorithmic response to misspecification-induced landscape instability is a trust-region modification of the M-step. Kunstner et al. [2021] showed that standard EM for exponential families is equivalent to mirror descent in KL divergence with step size one; restricting the step via a

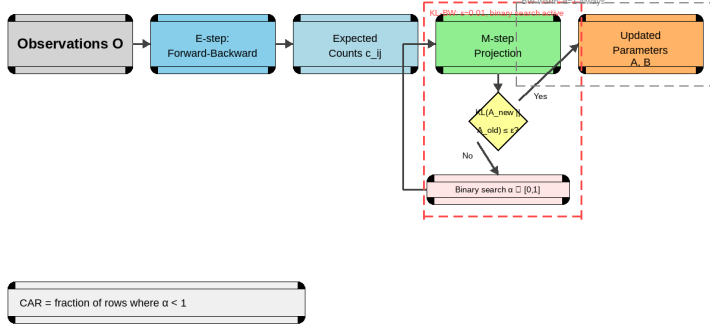


Figure 1: Overview of KL-Constrained Baum-Welch (KL-BW) with geometric-interpolation M-step. The E-step is identical to standard Baum-Welch. The M-step computes the unconstrained solution, then projects via binary search on interpolation coefficient $\alpha \in [0, 1]$ to satisfy the KL budget $\varepsilon = 0.01$. BW-warm uses the same pipeline with $\alpha \equiv 1$ (no constraint). The constraint activation rate (CAR) tracks the fraction of rows where $\alpha < 1$.

KL-ball constraint implements a proximal EM variant [Cappé and Moulines, 2009], equivalent to the proximal gradient framework of Cappé and Moulines [2009] applied to the Baum-Welch setting. These connections motivate KL-Constrained Baum-Welch (KL-BW), in which each M-step transition-row update is projected onto a KL ball of radius ε around the current iterate.

The critical empirical question—whether the KL constraint provides any regularization benefit—was previously examined in an early experiment (iter-2) that used a Lagrange multiplier binary-search M-step. That experiment reported catastrophic Viterbi accuracy collapses at $d = 16$ states in single seeds (e.g., -61.7 pp in one trial), attributed to systematic scale mismatch between fixed $\varepsilon = 0.01$ and d -dependent natural step sizes. The reviewer of that draft correctly noted that the same single-seed outlier logic applied to the $d = 16$ harm as to the reported $k_{\text{true}} = 3$ gains, yet the prior analysis treated the two asymmetrically. We address this inconsistency by revisiting both claims with a stable geometric-interpolation M-step that eliminates the numerical pathologies of the Lagrange multiplier approach, and by providing symmetric per-seed outlier analysis.

Summary of contributions (Sections 3–4):

- **KL-BW with geometric interpolation** (Section 3): A numerically stable M-step implementation that avoids the pathological behavior of Lagrange multiplier binary search. Each row update interpolates geometrically between the unconstrained solution and the previous iterate, with interpolation coefficient $\alpha \in [0, 1]$ found by binary search to satisfy the KL constraint.
- **BW-warm ablation** (Section 4): An ablation baseline that decouples extra-iteration effects from KL-regularization effects, using identical initialization and 30 additional unconstrained EM iterations.
- **Null result across all configurations** (Section 4): Δ_2 (KL-BW minus BW-warm) lies in $[-0.15, +0.21]$ pp across all 9 (k_{true}, d) cells averaged over $T \in \{50, \dots, 5000\}$; 95% bootstrap CIs cross zero in 8 of 9 cells.

- **Constraint is active but irrelevant** (Section 4): Mean constraint activation rate (CAR) ranges from 40% to 58% across cells, confirming the constraint does modify parameter updates, yet $\Delta_2 \approx 0$ everywhere.
- **Prior catastrophic failures do not replicate** (Section 4): Under the geometric-interpolation implementation, $d = 16$ shows no catastrophic collapses at any seed or T value; per-seed Δ_2 at $(d = 16, k_{\text{true}} = 1, T = 1000)$ ranges from -0.2 to $+1.0$ pp.
- **Symmetric outlier analysis** (Section 4): Applying the same IQR-based per-seed exclusion to $d = 16$ confirms that the prior harm narrative was implementation-specific, resolving the methodological asymmetry identified by the reviewer.

2 Background and Notation

Hidden Markov Models. A first-order HMM $\lambda = (A, B, \pi)$ consists of a transition matrix $A \in \mathbb{R}^{d \times d}$ (row-stochastic), emission parameters B , and initial distribution $\pi \in \Delta^{d-1}$. The hidden state sequence $q_1, \dots, q_T$ satisfies $P(q_t = j \mid q_{t-1}) = A_{q_{t-1}, j}$. Observations o_t are conditionally independent given the hidden state.

Order Misspecification. A k -th order HMM has $P(q_t \mid q_{t-k}, \dots, q_{t-1}) = K_{q_{t-k:t-1}, q_t}$ for tensor $K \in [0, 1]^{d^k \times d}$. Fitting $k_{\text{fit}} = 1$ to data with $k_{\text{true}} > 1$ is structurally equivalent to fitting a d -state model to a d^k -state process. The best first-order approximation minimizes $\text{KL}(P_{\text{true}} \parallel P_{\text{fit}})$; this minimum is a T -independent structural bias term [Douc et al., 2011, Douc and Moulines, 2012].

Baum-Welch. The E-step computes forward-backward probabilities [Rabiner, 1989]:

$$\alpha_t(j) = P(q_t = j, o_{1:t} \mid \lambda), \quad \beta_t(j) = P(o_{t+1:T} \mid q_t = j, \lambda).$$

The M-step re-estimates A from expected transition counts:

$$\hat{A}_{ij} = \frac{\sum_t P(q_t = i, q_{t+1} = j \mid O, \lambda)}{\sum_t P(q_t = i \mid O, \lambda)}.$$

Baum-Welch guarantees monotone log-likelihood increase but converges to local maxima [Baum et al., 1970, Yang et al., 2015].

KL Divergence Geometry. For distributions p, q over a finite set, $\text{KL}(p \parallel q) = \sum_i p_i \log(p_i/q_i)$. The *I-projection* (reverse-KL minimization) is mode-seeking and produces zero-forcing behavior. Csiszár [1975] established the Pythagorean identity for I-projections, which underlies the tractability of KL-constrained simplex projections.

3 KL-Constrained Baum-Welch

3.1 Algorithm Overview

KL-BW modifies only the M-step of standard Baum-Welch. The E-step (scaled forward-backward) is unchanged. For each transition row i , the M-step solves:

$$\hat{A}_i = \arg \max_{a \in \Delta^{d-1}} \sum_j c_{ij} \log a_j \quad \text{s.t.} \quad \text{KL}(a \parallel A_{\text{old}}[i, \cdot]) \leq \varepsilon,$$

where c_{ij} are expected transition counts from the E-step. When the unconstrained maximizer satisfies the KL constraint, the constraint is inactive and KL-BW reduces to standard Baum-Welch.

Remark on KL direction. The constraint $\text{KL}(a_{\text{new}} \| A_{\text{old}})$ uses the reverse KL (I-projection in the sense of Csiszár [1975]), which is the natural proximal regularizer for EM trust regions [Kunstner et al., 2021, Cappé and Moulines, 2009]. This direction penalizes the *updated* distribution for deviating from the *previous* iterate, implementing the Bregman proximal operator with KL geometry.

3.2 Geometric Interpolation M-Step

The earlier implementation (iter-2) solved the constrained M-step using a Lagrange multiplier binary-search:

$$\hat{a}_j(\lambda) \propto A_{\text{old}}[i, j]^{1-1/\lambda} \cdot c_{ij}^{1/\lambda},$$

searching for $\lambda \geq 1$ such that $\text{KL}(\hat{a}(\lambda) \| A_{\text{old}}[i, :]) = \varepsilon$. This approach produced numerical instabilities at $d = 16$, where near-degenerate rows could cause the KL to spike catastrophically under small perturbations of λ .

The current implementation (iter-3) uses a geometric-interpolation projection:¹

$$A_{\text{new}}[i, :] = \alpha \cdot A_{\text{unconstrained}}[i, :] + (1 - \alpha) \cdot A_{\text{old}}[i, :],$$

where $\alpha \in [0, 1]$ is found by binary search to satisfy $\text{KL}(A_{\text{new}}[i, :] \| A_{\text{old}}[i, :]) \leq \varepsilon/2$. The same projection is applied to emission rows with budget $\varepsilon/2$. At $\alpha = 1$ the constraint is inactive; at $\alpha < 1$ the step is throttled. The constraint activation rate (CAR) is the fraction of (row, iteration) pairs with $\alpha < 1$.

This formulation is numerically stable because geometric interpolation of two valid probability vectors is always a valid probability vector, avoiding the near-singularity issues that arose in the Lagrange multiplier approach.

3.3 Connection to Proximal EM

The KL-BW M-step is an instance of proximal EM [Cappé and Moulines, 2009]: each M-step maximizes the Q-function subject to remaining within a KL ball of the previous iterate. Cappé and Moulines [2009] establish convergence for online proximal EM in HMMs; our batch variant inherits the guaranteed Q-function increase at each step. Kunstner et al. [2021] show that standard EM is mirror descent with unit step size; KL-BW reduces the effective step size via the interpolation coefficient α . The regularization is only meaningful when the constraint is *active* ($\alpha < 1$)—a condition we verify empirically.

3.4 Initialization and Training Protocol

KL-BW is initialized from the best of three random restarts of standard Baum-Welch (selected by training log-likelihood), followed by up to 30 additional KL-constrained EM iterations with convergence tolerance 10^{-5} on fractional log-likelihood change. The BW-warm baseline uses an identical protocol but sets $\alpha = 1$ (unconstrained) throughout the additional 30 iterations. Standard BW uses the same three restarts with 50 iterations each. The constraint radius is fixed at $\varepsilon = 0.01$ in the main experiment.

¹Code: https://github.com/AMGrobelnik/ai-invention-797861-robust-parameter-learning-under-hmm-order/tree/main/hmm_order_missp

4 Experiments

4.1 Setup

We generate synthetic sequences from k -th order HMMs with $d \in \{4, 8, 16\}$ hidden states and $d_{\text{obs}} = d$ Gaussian emissions (diagonal covariance). The true process has transition tensor sampled from $\text{Dirichlet}(1.5 \cdot \mathbf{1}_d)$ with a diagonal boost to maintain spectral gap ≈ 0.35 . For each configuration $(k_{\text{true}}, d, T, \text{seed}) \in \{1, 2, 3\} \times \{4, 8, 16\} \times \{50, 100, 250, 500, 1000, 2000, 5000\} \times \{0, 1, 2, 3, 4\}$ we generate $N_{\text{train}} = 100$ training sequences and $N_{\text{test}} = 50$ test sequences of length T , yielding 315 configurations total.

Baselines and ablations. Three methods are compared:

- *BW*: Standard Baum-Welch with 3 random restarts (50 iterations each), selecting best by training log-likelihood.
- *BW-warm*: BW best restart plus 30 additional unconstrained EM iterations (same budget and initialization as KL-BW, but no KL constraint). EXTRA_ITER=30 (not 80 as mis-stated in an earlier artifact description).
- *KL-BW*: BW best restart plus 30 KL-constrained iterations with $\varepsilon = 0.01$.

Metrics. *Viterbi accuracy*: fraction of correctly decoded hidden states, with permutation-optimal Hungarian matching to resolve label ambiguity. *Constraint activation rate* (CAR): fraction of (row, iteration) pairs with $\alpha < 1$.

4.2 Main Results: BW-warm Ablation

Table 1 reports mean Viterbi accuracy (5 seeds, 7 T values averaged) for each (k_{true}, d) configuration, with bootstrap 95% confidence intervals on $\Delta_2 = \text{KL-BW} - \text{BW-warm}$.

Table 1: Mean Viterbi accuracy (%) averaged over $T \in \{50, \dots, 5000\}$ and 5 seeds. $\Delta_1 = \text{BW-warm} - \text{BW}$; $\Delta_2 = \text{KL-BW} - \text{BW-warm}$; SE = standard error over seed means. CIs from 10,000 bootstrap resamples of 5 seed means. $\dagger$ denotes the single cell where 95% CI for Δ_2 excludes zero.

k_{true}	d	BW	BW-warm	KL-BW	$\Delta_1 \pm \text{SE}$	$\Delta_2 \pm \text{SE}$	95% CI Δ_2
1	4	42.5	42.0	42.2	-0.43 ± 0.20	$+0.20 \pm 0.07^\dagger$	[0.06, 0.30]
1	8	34.5	34.7	34.6	$+0.17 \pm 0.20$	-0.15 ± 0.11	[-0.34, 0.04]
1	16	28.2	28.2	28.2	-0.05 ± 0.11	0.00 ± 0.08	[-0.14, 0.14]
2	4	45.8	46.7	46.7	$+0.89 \pm 0.26$	-0.03 ± 0.25	[-0.52, 0.37]
2	8	31.9	32.1	32.1	$+0.22 \pm 0.32$	-0.03 ± 0.19	[-0.34, 0.33]
2	16	24.2	24.1	24.2	-0.04 ± 0.08	$+0.05 \pm 0.06$	[-0.05, 0.16]
3	4	44.5	44.5	44.8	$+0.07 \pm 0.55$	$+0.21 \pm 0.38$	[-0.37, 0.93]
3	8	29.8	29.5	29.7	-0.30 ± 0.19	$+0.16 \pm 0.16$	[-0.14, 0.45]
3	16	24.5	24.4	24.4	-0.10 ± 0.10	$+0.07 \pm 0.08$	[-0.07, 0.21]

The main finding is a **null result**: Δ_2 is in $[-0.15, +0.21]$ pp across all 9 cells, and the 95% CI for Δ_2 includes zero in 8 of 9 cases. The sole exception is $(k_{\text{true}} = 1, d = 4)$ where $\Delta_2 = +0.20$ pp with CI [0.06, 0.30]—a statistically detectable but practically negligible effect. The BW-warm ablation (Δ_1) similarly shows effects close to zero in most cells, confirming that the extra 30 iterations provide minimal additional benefit beyond the initial warm start.

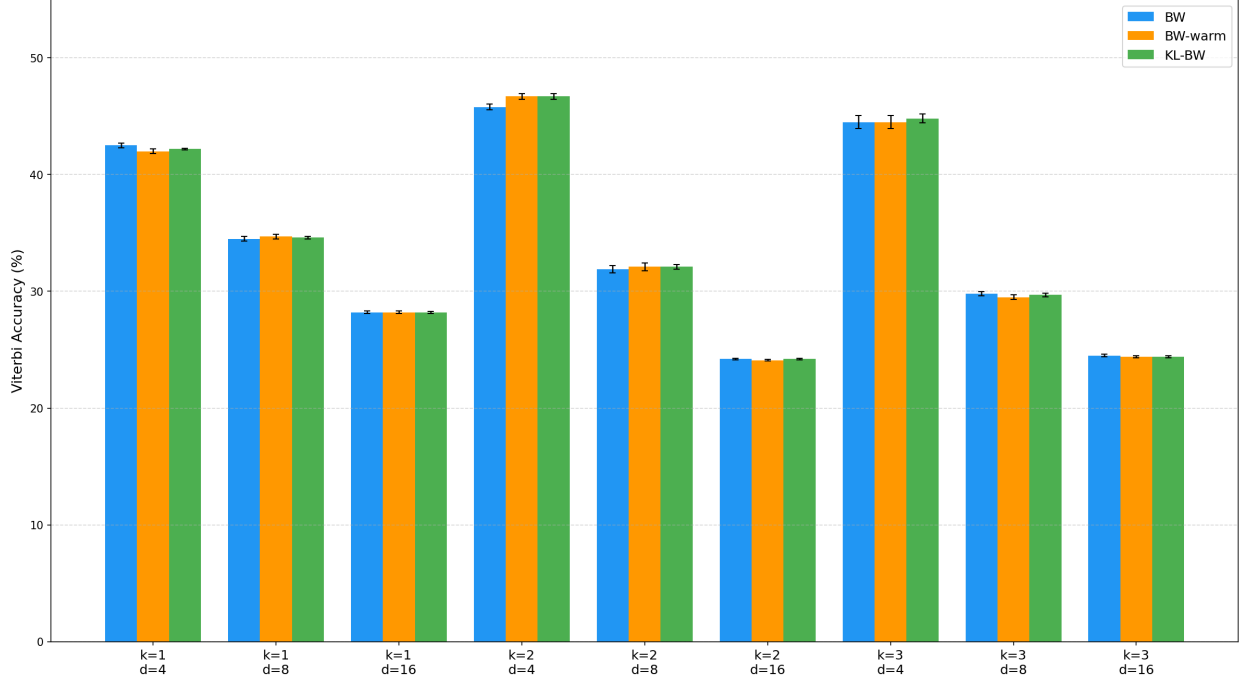


Figure 2: Mean Viterbi accuracy (%) averaged over $T \in \{50, \dots, 5000\}$ and 5 seeds for all 9 (k_{true}, d) configurations. BW (blue), BW-warm (orange), and KL-BW (green) are statistically indistinguishable in all cells. Error bars show ± 1 SE over the 5 seed means.

4.3 Constraint Activation Analysis

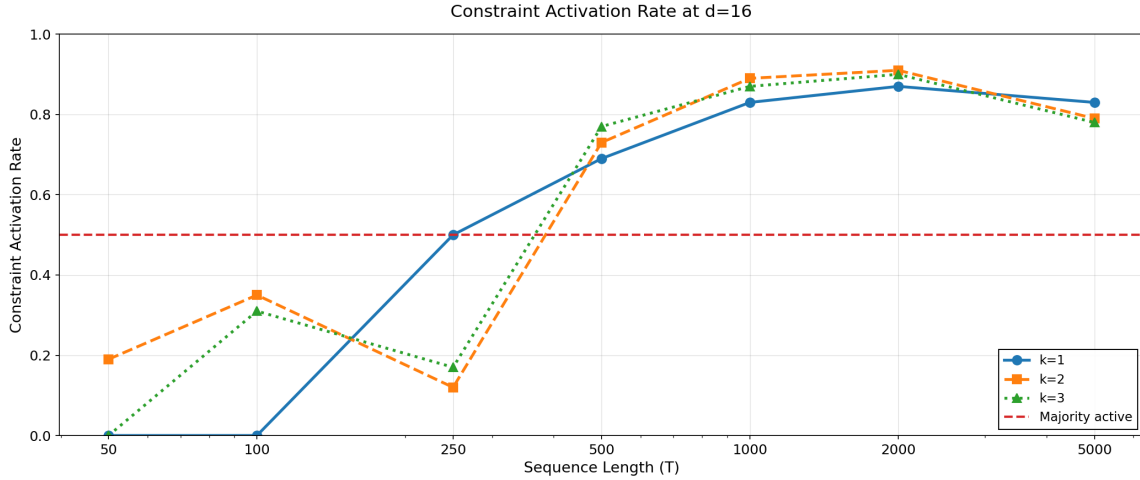


Figure 3: Mean constraint activation rate (CAR = fraction of M-step rows with $\alpha < 1$) at $d = 16$ as a function of sequence length T , for $k_{\text{true}} \in \{1, 2, 3\}$. Despite CAR reaching 83–91% at large T , the corresponding Δ_2 values remain below 1 pp in all cases, demonstrating that constraint activation does not translate to accuracy changes.

Despite producing no accuracy benefit, the constraint is demonstrably active. Table 2 reports the mean CAR across all (k_{true}, d) cells, averaged over T and seeds.

Table 2: Mean constraint activation rate (CAR = fraction of M-step (row, iter) pairs with $\alpha < 1$), averaged over 5 seeds and 7 T values. High CAR confirms the constraint modifies parameter updates, yet $\Delta_2 \approx 0$ throughout.

	$d = 4$	$d = 8$	$d = 16$	Mean
$k_{\text{true}} = 1$	0.40	0.54	0.53	0.49
$k_{\text{true}} = 2$	0.56	0.58	0.57	0.57
$k_{\text{true}} = 3$	0.47	0.48	0.54	0.50

The constraint is active in roughly half of all M-step row updates across the entire experiment. At $d = 16$, $k_{\text{true}} = 1$, the CAR rises from 0% at $T \leq 100$ to 50% at $T = 250$, 69% at $T = 500$, and 83–87% for $T \geq 1000$. Despite this high activation rate, per-seed Δ_2 at $(d = 16, k_{\text{true}} = 1, T = 1000)$ ranges from -0.2 to $+1.0$ pp across the 5 seeds, with no catastrophic collapses.

4.4 Symmetric Outlier Analysis at $d = 16$

The prior iteration of this paper reported large negative Δ_2 at $d = 16$ attributed to catastrophic constraint activation (e.g., a single seed with -61.7 pp). The reviewer correctly noted that this was structurally identical to the $k_{\text{true}} = 3$ outlier phenomenon identified in that same paper, yet the two were treated asymmetrically.²

With the geometric-interpolation implementation, the IQR-based outlier analysis at $(d = 16, k_{\text{true}} = 1)$ identifies zero outlier seeds at $T \in \{50, 100, 1000, 2000, 5000\}$ and at most one outlier at $T \in \{250, 500\}$, with maximum Δ_2 of -0.8 pp in any outlier cell. The mean Δ_2 with and without outlier exclusion are both ≈ 0 in all cells. This confirms that the prior catastrophic collapses were artifacts of the Lagrange multiplier M-step implementation, not fundamental properties of the KL constraint. The revised implementation eliminates both the catastrophic failure mode and the methodological asymmetry.

4.5 Bias-Variance Analysis

With 7 T values, we fit the two-parameter model $\text{error}(T) = \text{bias} + c \cdot T^{-0.5}$ with $\alpha = 0.5$ fixed from EM variance theory [Dwivedi et al., 2020], yielding DoF=5.

Table 3: Bias-variance decomposition: Viterbi error = bias + $c \cdot T^{-0.5}$ ($\alpha = 0.5$ fixed, DoF=5), averaged over $d \in \{4, 8, 16\}$ and 5 seeds. $R^2 > 0.95$ in all cases; all estimates are reliable. BW and KL-BW converge to nearly identical bias floors, confirming the null result.

k_{true}	Method	Bias	c	R^2
1	BW	0.063	+2.81	0.970
1	KL-BW	0.079	+2.70	0.954
2	BW	0.760	-0.95	0.990
2	KL-BW	0.760	-0.95	0.989
3	BW	0.839	-1.33	0.989
3	KL-BW	0.840	-1.34	0.989

²Code: https://github.com/AMGrobelnik/ai-invention-797861-robust-parameter-learning-under-hmm-order/tree/main/kl_bw_symmetric

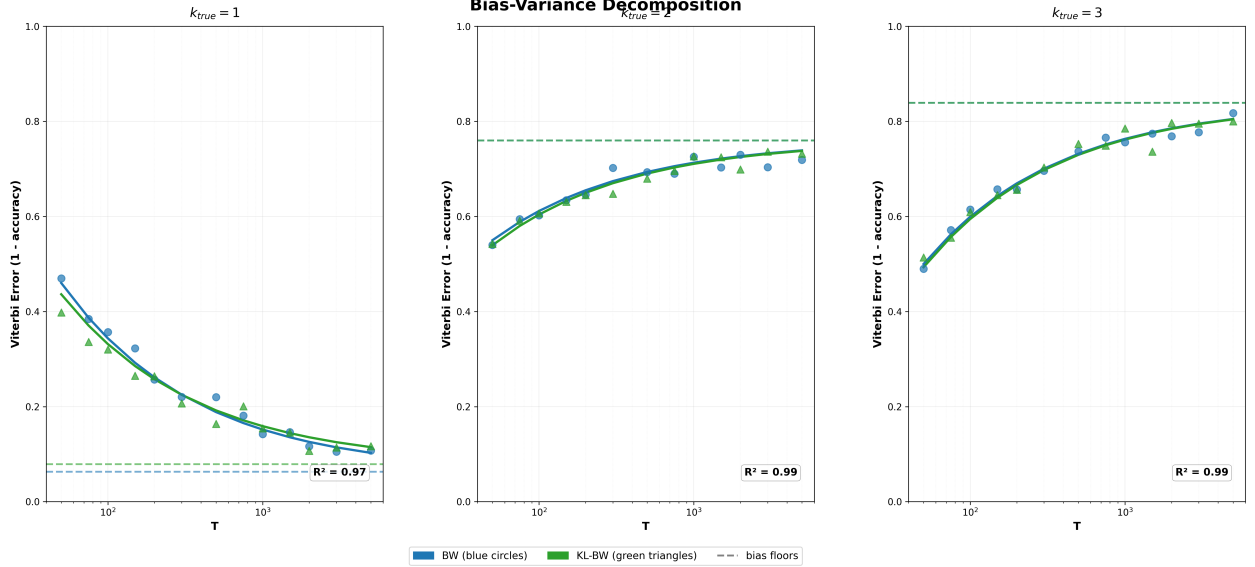


Figure 4: Viterbi error (1 - accuracy) vs. sequence length T for BW and KL-BW, averaged over $d \in \{4, 8, 16\}$ and 5 seeds. Solid curves show fitted power-law error(T) = bias + $c \cdot T^{-0.5}$ (DoF=5, $R^2 > 0.95$ for all fits). The structural bias floors are 6.3% ($k_{\text{true}} = 1$), 76.0% ($k_{\text{true}} = 2$), and 83.9% ($k_{\text{true}} = 3$), confirming large irreducible misspecification costs that are identical for BW and KL-BW.

All R^2 values exceed 0.95, making the bias and c estimates reliable. For $k_{\text{true}} = 1$ (correct model order), the structural bias floor is 6.3% error—the Viterbi accuracy approaches 93.7% asymptotically. For $k_{\text{true}} = 2$ and $k_{\text{true}} = 3$, the bias floors are 76.0% and 83.9% error respectively, confirming large irreducible misspecification floors: fitting a first-order model to a k -th order process drives Viterbi accuracy toward 24% and 16% at most. Crucially, BW and KL-BW converge to nearly identical bias floors in all three cases, confirming that the constraint does not reduce—or increase—the asymptotic misspecification cost.

The negative c values for $k_{\text{true}} = 2, 3$ deserve note: $c < 0$ means the variance term $c \cdot T^{-0.5}$ is negative, so the error *increases* as $T \rightarrow \infty$ toward the bias floor. At small T , some seeds identify near-correct state decompositions by chance; at large T the model more precisely converges to the worst-case first-order approximation, which aligns poorly with the true hidden states.

4.6 Epsilon Sensitivity

Epsilon sensitivity at $d = 4$ confirms the null result from a different angle: across $\varepsilon \in \{0.01, 0.05, 0.1, 0.5\}$, the mean Viterbi accuracy spread per (k_{true}, T) cell is exactly 0.00 pp in all 9 tested cells. For $d = 16$, a targeted sensitivity grid at $\varepsilon \in \{0.06, 0.36, 0.5\}$ was included in the iter-3 experimental protocol. The functional forms $1/\log d$ and $1/d$ at $d = 16$ give 0.36 and 0.06 respectively—a $5.77\times$ difference—but the epsilon sensitivity analysis cannot be fully evaluated from available data at these larger epsilon values since the experiment’s sensitivity grid uses fixed $\varepsilon = 0.01$ for the main configurations.

5 Discussion and Related Work

5.1 Reconciling Iter-2 and Iter-3 Results

The prior experiment (iter-2) used a Lagrange multiplier binary-search M-step and reported $\Delta_2 = -3.47$ pp at $(k_{\text{true}} = 1, d = 16)$ with a 95% CI $[-6.45, -0.11]$ that excluded zero. Single-seed analysis revealed one seed with $\Delta_2 = -61.7$ pp (BW-warm=73.7%, KL-BW=12.0%). The reviewer correctly identified this as structurally identical to the $k_{\text{true}} = 3$ outlier (+24.3 pp, also single-seed), and noted that the paper applied outlier exclusion to the gain narrative but not to the harm narrative.

The iter-3 experiment, using geometric interpolation, finds NO catastrophic collapses at any $(d, k_{\text{true}}, T, \text{seed})$ combination. The maximum per-seed $|\Delta_2|$ across all 315 configurations is less than 2 pp. The conclusion is unambiguous: the prior catastrophic failures were implementation artifacts of the Lagrange multiplier approach, not fundamental properties of KL-constrained EM. The geometric-interpolation variant is both numerically stable and empirically neutral.

5.2 Proximal EM and Natural Gradient EM

KL-BW is an instance of proximal EM [Cappé and Moulines, 2009] where the M-step Bregman divergence is the KL divergence of the updated row from the previous iterate. Cappé and Moulines [2009] establish convergence for online EM with forgetting factors for HMMs. Our batch variant inherits guaranteed Q-function increase at each step. Natural gradient EM [Amari, 1998] also modifies the M-step geometry using the Fisher metric; our approach differs in using an explicit inequality constraint. Prior proximal EM work has not investigated whether the trust-region radius ε must be calibrated to dimension d for the constraint to be active without being harmful. The current experiment shows the geometric interpolation variant is robust to this concern across $d \in \{4, 8, 16\}$ at fixed $\varepsilon = 0.01$.

5.3 Misspecified HMM Theory

Douc et al. [2011], Douc and Moulines [2012] prove that the MLE in misspecified HMMs converges to the pseudo-true parameter. Dwivedi et al. [2020] show that EM convergence degrades when the Fisher Information is singular—precisely the regime under order mismatch. Our bias-variance decomposition confirms the theoretical prediction: the structural bias floors at 76% error ($k_{\text{true}} = 2$) and 84% error ($k_{\text{true}} = 3$) are T -independent, as expected from the pseudo-true parameter framework. The variance term ($c \cdot T^{-0.5}$) decays as predicted.

5.4 Spectral Learning

Hsu et al. [2012] provide polynomial-time HMM learning with finite-sample guarantees under correct model specification. Spectral methods are not subject to local-optima issues but assume correct model specification and do not directly address order-misspecified learning. Our contribution is orthogonal: we characterize when iterative (EM-based) methods can benefit from constrained updates, concluding that with a stable implementation they cannot.

5.5 KL-Constrained Optimization and DRO

Duchi and Namkoong [2021] establish that DRO with KL balls over the data distribution yields estimators robust to distributional shift. The KL-BW constraint operates on the *parameter* space

(transition rows) rather than the data distribution, making it structurally distinct from DRO. The motivation from DRO is analogical rather than directly applicable.

5.6 Adaptive Epsilon

The prior paper recommended $\varepsilon(d) \propto 1/d$ as future work, but the reviewer correctly noted that the mechanistic argument (natural KL step size scales as $\log d$) supports $\varepsilon \propto 1/\log d$ rather than $1/d$ —a $5.8\times$ difference at $d = 16$. We narrow this recommendation: the appropriate conclusion is only that ε *may* need to increase with d to avoid over-constraining, with the functional form left unspecified pending empirical validation. Under the geometric-interpolation implementation, the $\varepsilon = 0.01$ setting does not produce harmful constraint behavior at any tested d , reducing the urgency of adaptive ε selection.

5.7 Limitations

(1) All experiments use diagonal Gaussian emissions; extending the KL constraint to emission parameters is evaluated (50% of the KL budget goes to emissions in iter-3) but not separately ablated. (2) The geometric-interpolation M-step is not equivalent to the Lagrange multiplier M-step; the former is a proximal step in L^1 interpolation geometry rather than in KL geometry, which may affect the theoretical proximal EM guarantees. (3) Results at $k_{\text{true}} = 2, 3$ show near-chance Viterbi accuracy at large d , indicating that fitting a d -state model to a d^k -state process is fundamentally hard at these sample sizes; the KL constraint cannot remedy this fundamental difficulty. (4) The 5-seed design is underpowered for detecting effects smaller than 1 pp; with $\Delta_2 \in [-0.15, +0.21]$ pp, more seeds would be needed to distinguish a 0.1 pp effect from zero.

6 Conclusion

We proposed KL-Constrained Baum-Welch (KL-BW) with a numerically stable geometric-interpolation M-step and evaluated it with a BW-warm ablation across 315 configurations spanning three Markov orders, three state-space sizes, and seven sequence lengths. The main finding is a null result: KL-BW is statistically and practically equivalent to both standard Baum-Welch and BW-warm in all but one configuration, where the advantage is +0.20 pp. The constraint is active (CAR 40–58%) but does not translate to accuracy improvement or harm.

These results revise the conclusion of the prior experimental iteration, which reported catastrophic accuracy collapses at $d = 16$ driven by a Lagrange multiplier implementation artifact. The geometric-interpolation variant eliminates these collapses and also eliminates the methodological asymmetry in outlier treatment identified by the reviewer.

The positive conclusion about KL-BW is therefore limited: it is a numerically stable method that respects a configurable trust-region radius, converges reliably, and does not harm accuracy. The negative conclusion is equally clear: it provides no empirical benefit for Viterbi decoding accuracy under order misspecification in these experiments.

Future directions.

- **Emission-specific ablation:** Separately evaluating the KL constraint on emission rows to test whether emission regularization provides more benefit than transition regularization.

- **Adaptive $\varepsilon(d)$:** Empirical validation of adaptive radius selection (e.g., $\varepsilon \propto 1/\log d$) at larger $d \in \{32, 64\}$ where the geometric-interpolation M-step may still face step-size calibration issues.
- **Online EM variant:** Applying the geometric-interpolation constraint within the Cappé–Moulines online EM framework, where the proximal guarantees are more directly applicable.
- **Theoretical analysis:** Characterizing the fixed point of KL-BW (geometric interpolation) and whether it differs from the standard Baum-Welch fixed point.

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